

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-16

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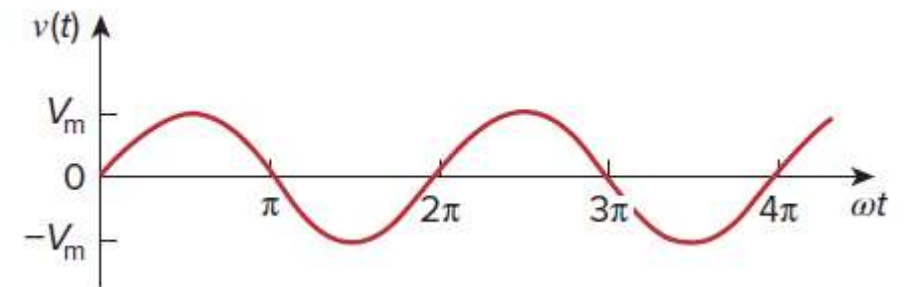
II.RMS(Root Mean Square) Value:

- Effective Value/True Value/AC Circuit Value
- It is the equivalent steady value of time varying waveform which could transfer the same amount of energy/power/heat as given by the original waveform for the definite period of time.
- The effective value of periodic current is root mean square value

$$I_{ef} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} = I_{rms}$$

- Similarly the effective value of periodic voltage is

$$V_{ef} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} = V_{rms}$$



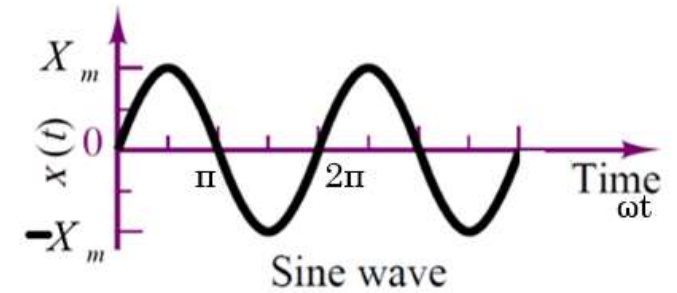
Effective or RMS Values

- The effective value of periodic current is root mean square value

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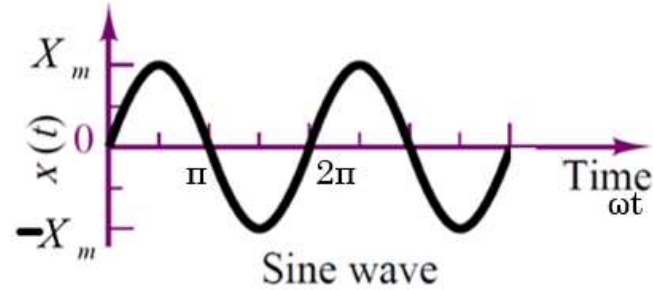
Form factor

- It is the ratio between RMS value to the Average value.
- It signifies the required AC voltage or current to do the same effect as it is in the DC voltage or current.
- The magnitude in the AC equivalent varies with respect to the shape of the wave form.

$$\text{Form factor} = \frac{\text{RMS value}}{\text{Average value}}$$

1. Find the RMS value Sinusoidal Wave

1.RMS value of Sinusoidal Wave



$$X_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} x^2(\omega t) d\omega t}$$

$$X_{rms} = \sqrt{\frac{X_m^2}{4\pi} \int_0^{2\pi} (1 - \cos 2\omega t) d\omega t}$$

$$X_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} X_m^2 (\sin^2 \omega t) d\omega t}$$

$$X_{rms} = \sqrt{\frac{X_m^2}{4\pi} \left(\omega t - \frac{\sin 2\omega t}{2} \right)_0^{2\pi}}$$

$$X_{rms} = \sqrt{\frac{X_m^2}{2\pi} \int_0^{2\pi} \left(\frac{1 - \cos 2\omega t}{2} \right) d\omega t}$$

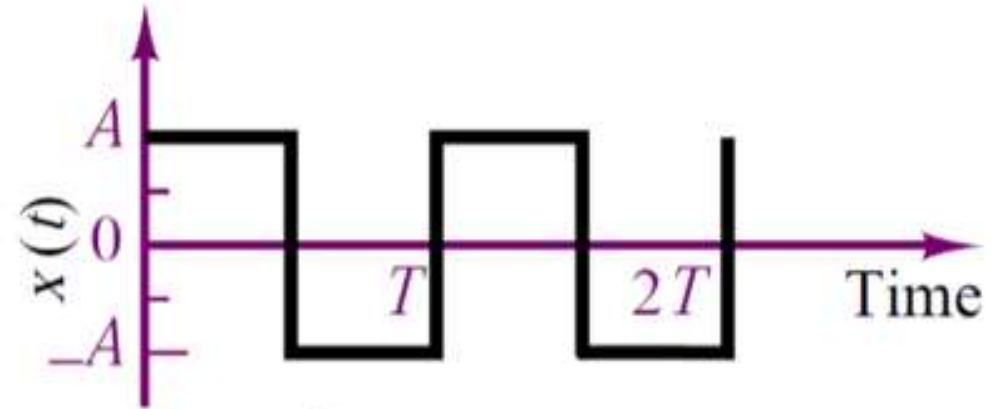
$$X_{rms} = \sqrt{\frac{X_m^2}{4\pi} ((2\pi - 0) - 0)} = \frac{X_m}{\sqrt{2}}$$

Form Factor

$$\begin{aligned}\text{Form factor} &= \frac{X_{rms}}{X_{av}} \\ &= \frac{X_m / \sqrt{2}}{2 X_m / \pi} \\ &= \frac{\pi}{2\sqrt{2}} = 1.11\end{aligned}$$

2. Find the RMS value of Square Wave

2. Square wave



Square wave

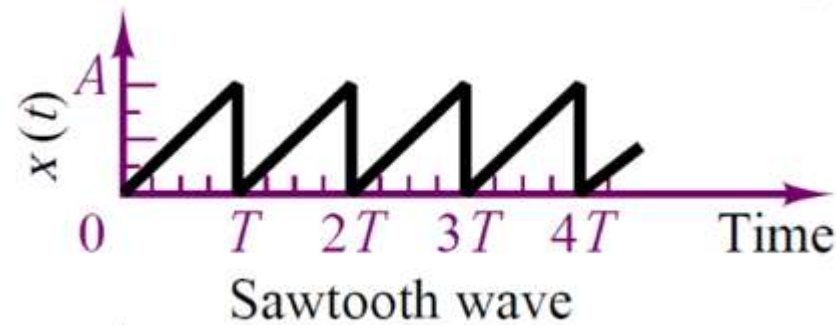
$$X_{av} = \frac{A \times T/2}{T/2} = A$$

$$X_{rms} = \sqrt{\frac{A^2 \times T/2}{T/2}} = A$$

$$\text{Form factor} = \frac{X_{rms}}{X_{av}} = \frac{A}{A} = 1$$

3. Find the RMS value of Sawtooth Wave

3. Sawtooth Wave:

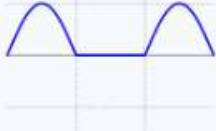
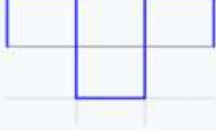
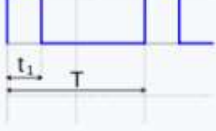


$$X_{rms} = \sqrt{\frac{\frac{1}{3} \times A^2 \times T}{T}} = \frac{A}{\sqrt{3}}$$

$$X_{av} = \frac{\left(\frac{1}{2} \times A \times T \right)}{T} = \frac{A}{2}$$

$$\text{Form factor} = \frac{X_{rms}}{X_{av}} = \frac{A/\sqrt{3}}{A/2} = \frac{2}{\sqrt{3}}$$

Find the RMS value of certain periodic signals

Waveform	Image	RMS	ARV	Form Factor
Sine wave		$\frac{a}{\sqrt{2}}^{[2]}$	$a \frac{2}{\pi}^{[2]}$	$\frac{\pi}{2\sqrt{2}} \approx 1.11072073^{[3]}$
Half-wave rectified sine		$\frac{a}{2}$	$\frac{a}{\pi}$	$\frac{\pi}{2} \approx 1.571$
Full-wave rectified sine		$\frac{a}{\sqrt{2}}$	$a \frac{2}{\pi}$	$\frac{\pi}{2\sqrt{2}}$
Square wave, constant value		a	a	$\frac{a}{a} = 1$
Pulse wave		$a\sqrt{D}^{[6]}$	aD	$\frac{1}{\sqrt{D}} = \sqrt{\frac{T}{\tau}}$
Triangle wave		$\frac{a}{\sqrt{3}}^{[7]}$	$\frac{a}{2}$	$\frac{2}{\sqrt{3}} \approx 1.15470054$
Sawtooth wave		$\frac{a}{\sqrt{3}}$	$\frac{a}{2}$	$\frac{2}{\sqrt{3}}$